

Program on strings in CUDA

Name: Satyam Singh

Reg no: 230905256

CSE B2 - 37

Sample Question:

A CUDA program which takes a string as input and determines the number of occurrences of a character 'a' in the string. This program uses atomicAdd() function.

Ans)

```
#include "cuda_runtime.h" #include  
"device_launch_parameters.h"
```

```
#include <stdio.h> #include <stdlib.h> #include <string.h>
```

```
#define N 1024
```

```
global void CUDACount(char* A, unsigned int *d_count){ int i =  
threadIdx.x; if (A[i] == 'a') atomicAdd(d_count,1); }
```

```
int main() { char A[N]; char *d_A;
```

```
unsigned int *count = (unsigned int *) malloc(sizeof(unsigned  
int));
```

```
unsigned int *d_count;
unsigned int *result = (unsigned int *) malloc(sizeof( unsigned
int));

printf("Enter a string : ");

fgets(A, N, stdin);
A[strcspn(A, "\n")] = 0;

cudaEvent_t start, stop;
cudaEventCreate(&start);
cudaEventCreate(&stop);
cudaError_t error =cudaGetLastError();

cudaMalloc((void**)&d_A, strlen(A)*sizeof(char));
cudaMalloc((void**)&d_count, sizeof(unsigned int));

cudaMemcpy(d_count, count, sizeof(unsigned int),
cudaMemcpyHostToDevice);
cudaMemcpy(d_A, A, strlen(A), cudaMemcpyHostToDevice);

*count = 0;

if (error != cudaSuccess)
    printf("CUDA Error!: %s\n", cudaGetErrorString(error));

cudaEventRecord(start, 0);
```

```

CUDACount<<<1,strlen(A)>>>(d_A,d_count);
cudaDeviceSynchronize();

cudaEventRecord(stop, 0);
cudaEventSynchronize(stop);

error = cudaGetLastError();
if (error != cudaSuccess)
    printf("CUDA Error2: %s\n", cudaGetErrorString(error));

float elapsedTime;

cudaEventElapsedTime(&elapsedTime, start, stop);
cudaMemcpy(result, d_count, sizeof(unsigned int),
cudaMemcpyDeviceToHost);

printf("Total occurrences of a = %u\n", *result);
printf("Time Taken = %f\n",elapsedTime);

cudaFree(d_A);
cudaFree(d_count);
printf("\n");
getchar();
return 0;

}

```

Output:

```
STUDENT@MIT-ICT-LAB5-03:~/Desktop/230905256/Lab7$ nvcc sample.cu -o sample
STUDENT@MIT-ICT-LAB5-03:~/Desktop/230905256/Lab7$ ./sample
Enter a string : This is CUDA programming
Total occurrences of a = 1
Time Taken = 0.013152
```

Lab Exercises:

Q1)

Write a program in CUDA to count the number of times a given word is repeated in a sentence. (Use Atomic function)

Ans)

```
#include "cuda_runtime.h"
```

```
#include "device_launch_parameters.h"
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#define N 1024
```

```
__global__ void CUDACountWord(char* A, char* word, unsigned int  
*d_count, int strLen, int wordLen){
```

```
int i = threadIdx.x;

if (i <= strLen - wordLen) {
    int match = 1;

    for (int j = 0; j < wordLen; j++) {
        if (A[i + j] != word[j]) {
            match = 0;
            break;
        }
    }

    if (match) {
        atomicAdd(d_count, 1);
    }
}

int main() {
    char A[N], word[50];
    char *d_A, *d_word;
```

```
unsigned int *count = (unsigned int *) malloc(sizeof(unsigned int));
unsigned int *d_count;
unsigned int *result = (unsigned int *) malloc(sizeof(unsigned int));

printf("Enter a string : ");
fgets(A, N, stdin);
A[strcspn(A, "\n")] = 0;

printf("Enter word to search : ");
fgets(word, 50, stdin);
word[strcspn(word, "\n")] = 0;

int strLen = strlen(A);
int wordLen = strlen(word);

cudaEvent_t start, stop;
cudaEventCreate(&start);
cudaEventCreate(&stop);

cudaMalloc((void**)&d_A, strLen * sizeof(char));
```

```
cudaMalloc((void**)&d_word, wordLen * sizeof(char));
cudaMalloc((void**)&d_count, sizeof(unsigned int));

*count = 0;

cudaMemcpy(d_count, count, sizeof(unsigned int),
cudaMemcpyHostToDevice);
cudaMemcpy(d_A, A, strLen, cudaMemcpyHostToDevice);
cudaMemcpy(d_word, word, wordLen, cudaMemcpyHostToDevice);

cudaEventRecord(start, 0);

CUDACountWord<<<1, strLen>>>(d_A, d_word, d_count, strLen,
wordLen);
cudaDeviceSynchronize();

cudaEventRecord(stop, 0);
cudaEventSynchronize(stop);

float elapsedTime;
cudaEventElapsedTime(&elapsedTime, start, stop);
```

```
cudaMemcpy(result, d_count, sizeof(unsigned int),
cudaMemcpyDeviceToHost);

printf("Total occurrences of \"%s\" = %u\n", word, *result);
printf("Time Taken = %f ms\n", elapsedTime);

cudaFree(d_A);
cudaFree(d_word);
cudaFree(d_count);

printf("\n");
getchar();
return 0;
}
```

Output:

```
● STUDENT@MIT-ICT-LAB5-03:~/Desktop/230905256/Lab7$ nvcc q1.cu -o q1
○ STUDENT@MIT-ICT-LAB5-03:~/Desktop/230905256/Lab7$ ./q1
Enter a string : Hello Hai Hai See you
Enter word to search : Hai
Total occurrences of "Hai" = 2
Time Taken = 0.013312 ms
```

Q2) Write a CUDA program that reads a string S and produces the string RS as follows:

input string S : PCAP

Output string RS : PCAPPCAPCP

Note:Each work item copies required number of characters from S in RS.

Ans)

```
#include "cuda_runtime.h"
```

```
#include "device_launch_parameters.h"
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define N 1024
```

```
__global__ void buildString(char *S, char *RS, int len) {
```

```
int i = threadIdx.x;

if (i < len) {
    int start = 0;

    // Calculate starting index in RS
    for (int k = 0; k < i; k++) {
        start += (len - k);
    }

    // Copy (len - i) characters
    for (int j = 0; j < (len - i); j++) {
        RS[start + j] = S[j];
    }
}

int main() {
    char S[N];
    char *d_S, *d_RS;
```

```
printf("Enter a string: ");  
fgets(S, N, stdin);  
S[strcspn(S, "\n")] = 0;  
  
int len = strlen(S);  
  
// Total size of RS = len + (len-1) + ... + 1  
int totalLen = len * (len + 1) / 2;  
  
char *RS = (char*) malloc(totalLen + 1);  
  
cudaMalloc((void**)&d_S, len * sizeof(char));  
cudaMalloc((void**)&d_RS, totalLen * sizeof(char));  
  
cudaMemcpy(d_S, S, len, cudaMemcpyHostToDevice);  
  
buildString<<<1, len>>>(d_S, d_RS, len);  
cudaDeviceSynchronize();  
  
cudaMemcpy(RS, d_RS, totalLen, cudaMemcpyDeviceToHost);
```

```
RS[totalLen] = '\0';
```

```
printf("Output string RS: %s\n", RS);
```

```
cudaFree(d_S);
```

```
cudaFree(d_RS);
```

```
free(RS);
```

```
return 0;
```

```
}
```

Output:

```
STUDENT@MIT-ICT-LAB5-03:~/Desktop/230905256/Lab7$ nvcc q2.cu -o q2
STUDENT@MIT-ICT-LAB5-03:~/Desktop/230905256/Lab7$ ./q2
Enter a string: PCAP
Output string RS: PCAPPCAPCP
```